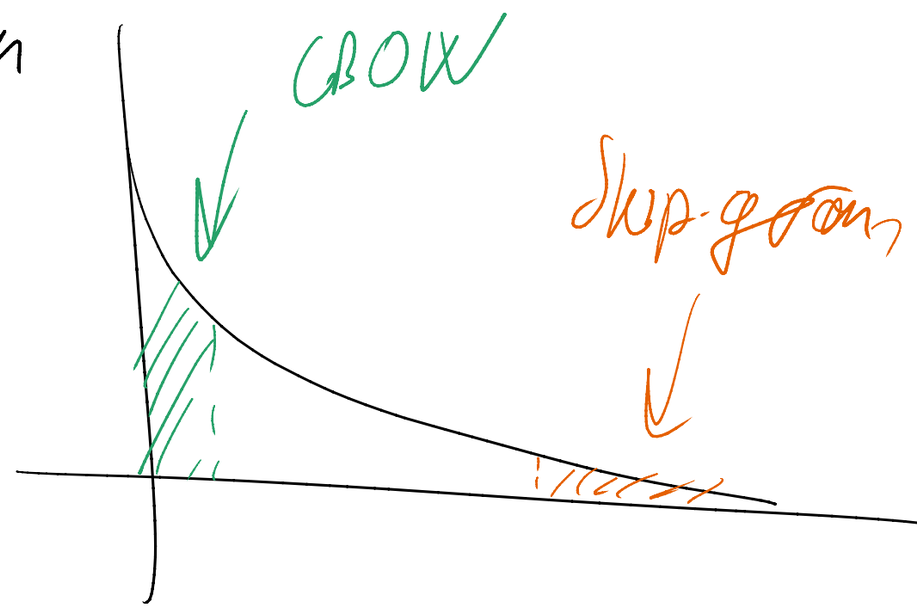
$V, V^T, \frac{1}{2}(V + V^T)$

$V_{man} - V_{king} \approx V_{woman} - V_{queen}$

neg sampling



CBOW frequent

Skip-gram rare

Fasttext (***)

W2V (Google)

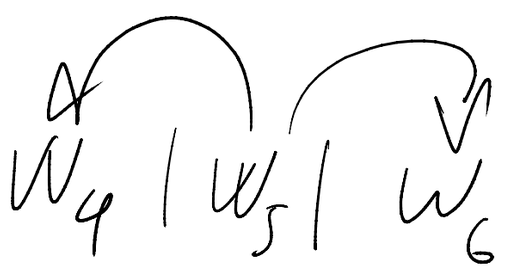
$p(добрый | хороший)$

$p(добрый | хо)$

$| хор$

$| орш$

$| ...$



$p(w_4 | w_5)$ . .

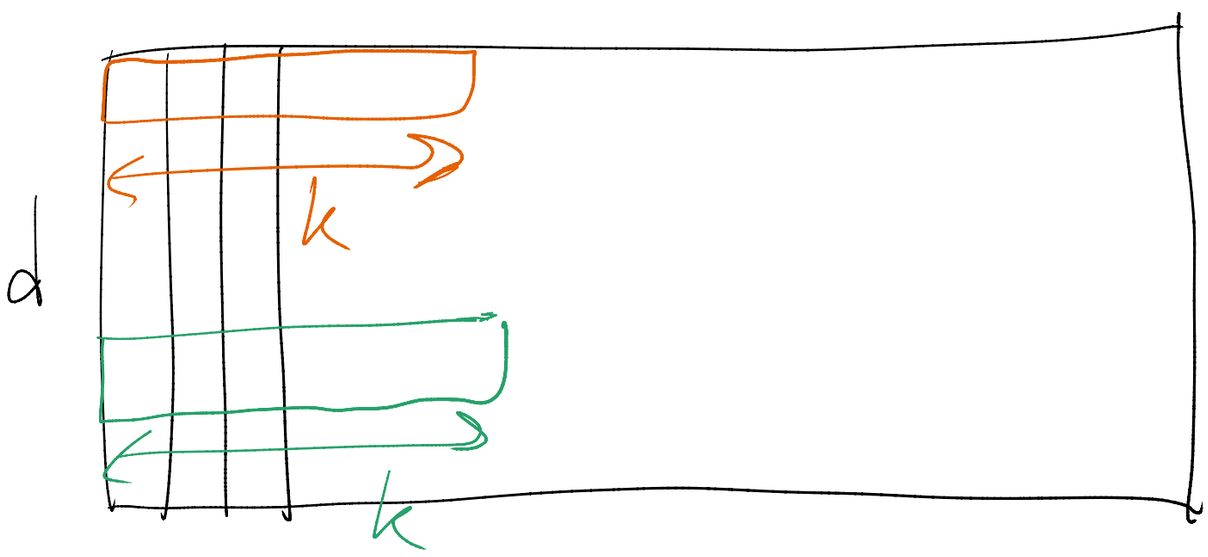
n-gram

$n = 3 \ldots 6$

+ sum of n-gram embeds — new embed
+ info exchange between words

Glove (Global vectors)

Text CNN



1-d conv

d - channels

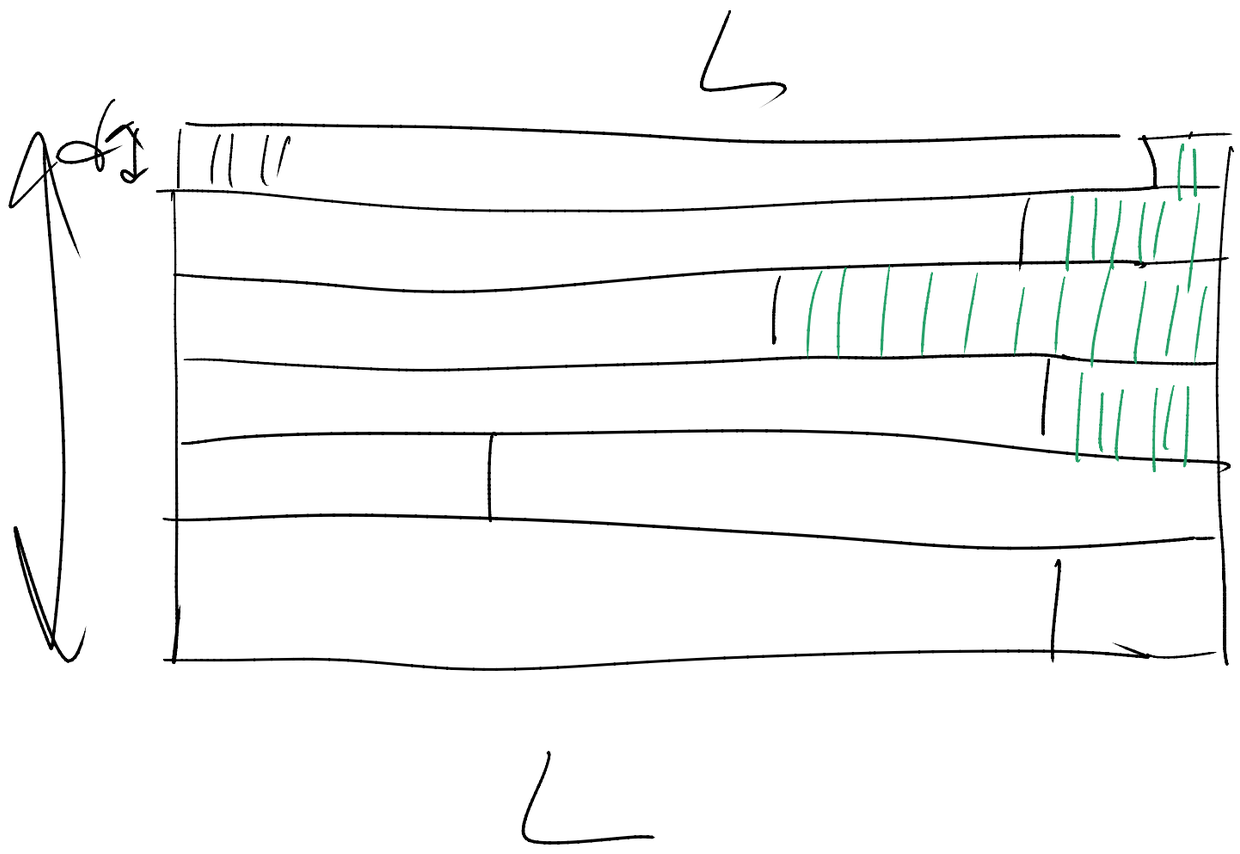
L - spatial dimension

L

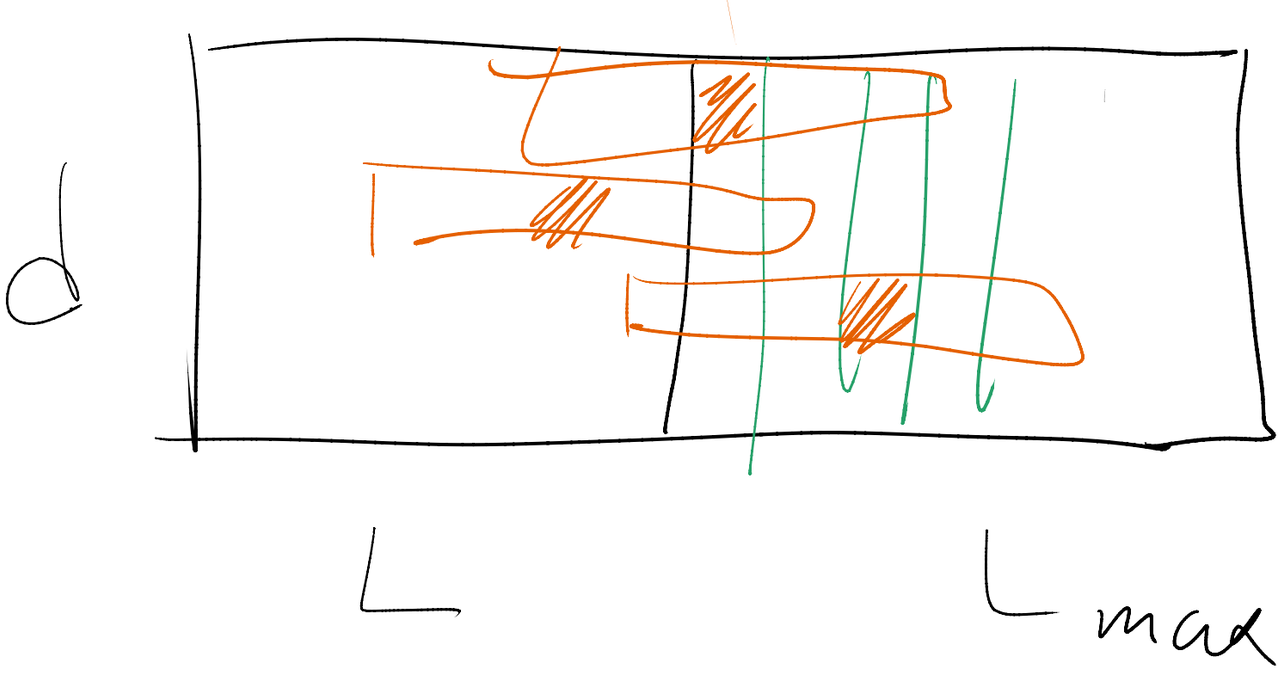
d × d' , k × 1

B × C × L

B × L × C



<PAD> - padding token



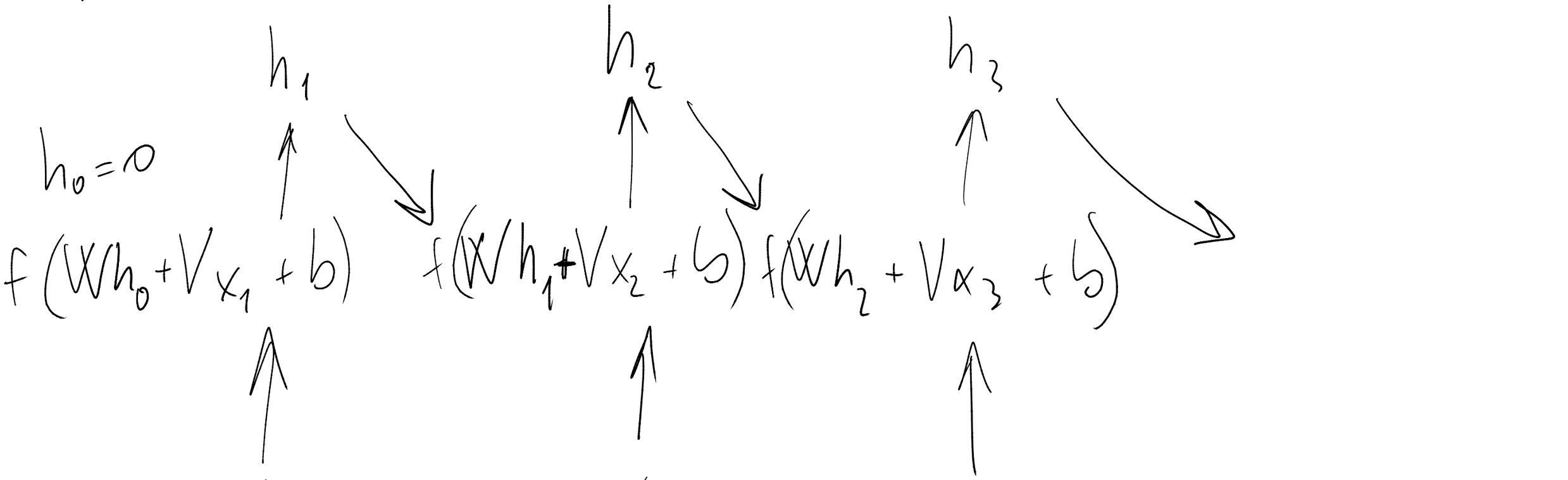
Recurrent Neural Networks

$x_1, t_2, t_3 \dots t_L$

$p(T_3 | t_1, t_2)$

$p(T_i | t_1 \dots t_{i-1})$

$p(T_i | t_{i-1}) \cdot p(t_{i-1} | t_1 .. t_{i-2})$



x_{t-2}

- $\mathcal{L} = \frac{1}{L} \sum_{t=1}^{L} Q(y_{true}, y_t)$

- $\frac{d\mathcal{L}}{dU} = \frac{1}{L} \sum_{t=1}^{L} \frac{dQ}{dy_t} \cdot \frac{dy_t}{dU}$

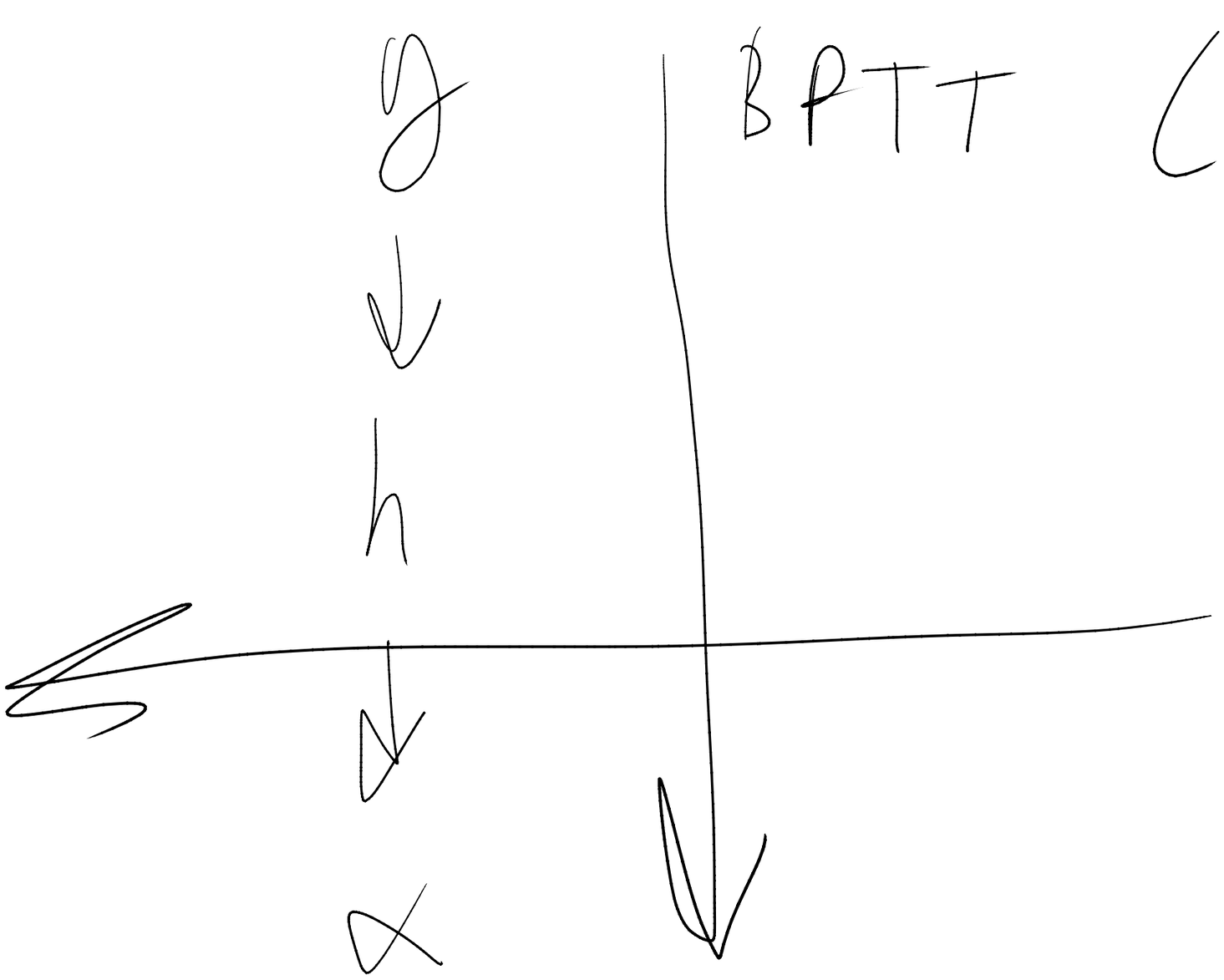
- $\frac{d\mathcal{L}}{dW} = \frac{1}{L} \sum_{t=1}^{L} \frac{dQ}{dy_t} \frac{dy_t}{dW} \frac{dh_t}{dW}$ =

$\frac{dh_t}{dW} = \frac{\partial h_t}{\partial W} + \frac{\partial h_t}{\partial h_{t-1}} \cdot \frac{dh_{t-1}}{dW} =$

$= \sum_{j=0}^{t-1} \left(\prod_{k=t-j}^{t-1} \frac{\partial h_{k+1}}{\partial h_k} \right) \cdot \frac{dh_{t-j}}{dW}$

$\frac{dh_t}{\partial W} + \frac{dh_t}{\partial h_{t-1}} \cdot \left(\frac{\partial h_{t-1}}{\partial W} + \frac{\partial h_{t-1}}{\partial h_{t-2}} \cdot \frac{dh_{t-2}}{dW} \right)$

$$= \frac{1}{L} \sum_{t=1}^{L} \frac{dQ}{dy_t} \frac{dy_t}{dh_t} \left[\sum_{j=0}^{t-1} \left(\prod_{k=t-j}^{t-1} \frac{\partial h_{k+1}}{\partial h_k} \right) \frac{dh_{t-j}}{dw} \right]$$



BPTT (Backprop through time)

Autoreggressive modeling (Language Model)

$$p(x) = p(x_1)\, p(x_2 | x_1) \cdot p(x_3 | x_1, x_2) \cdots p(x_L | x_1 \cdots x_{L-1})$$